# Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardise
- Step 9: Summarise & Plan future actions
- Guidelines to user

# Concern Resolution Process for In-house Manufacturing

# **QC Story**

# QC STORY PROJECT

**PROJECT TITLE :- Elimination of Damage Rocker Panel**

**Mahindra and Mahindra Ltd,__Nasik___ Plant, _____I__**
**.**

| | Plan | Actual |
|---|---|---|
| Start Date | **20.05.12** | **20.05.12** |
| Completion Date | **20.09.12** | **20.09.12** |

**Current Status:**

**CLOSE**

# Cross Functional Team

Team Leader – Leads the team, Co-ordinates the meeting
Mentor - Facilitates the team for concern resolution, provides resource.
Mostly Mentor is PU Manager.
Members – Cross Functional Team

| No | Description | Name | Designation | Dept | Phone | Email |
|---|---|---|---|---|---|---|
| 1 | **Team Leader** | Dinesh | L7 M | Production | 9850848982 | Jangam.dinesh@mahindra.com |
| 2 | **Mentor** | R Suryavanshi | L4 DGM | HOD | 9881378988 | Suryavanshi.ravindra@mahindra.com |
| 3 | **Member 1** | S Dwivedi | L8 O | Tooling | 9922971514 | Dwivedi.sushil2@mahindra.com |
| 4 | **Member 2** | N Hiray | L7 O | Quality | 9881737077 | Hiray.nitin2@mahindra.com |
| 5 | **Member 3** | A Hasme | L6 M | Tooling | 9822248250 | Hasme.ashutosh@mahindra.com |
| 6 | **Member 4** | R Dhanokar | L7 M | ID | 9881250515 | Dhanokar.rajesh@mahindra.com |
| 7 | **Member 5** | | | | | |

# 1. Select the Theme

## Brief description of Problem

WHAT - Rocker Panel damage.

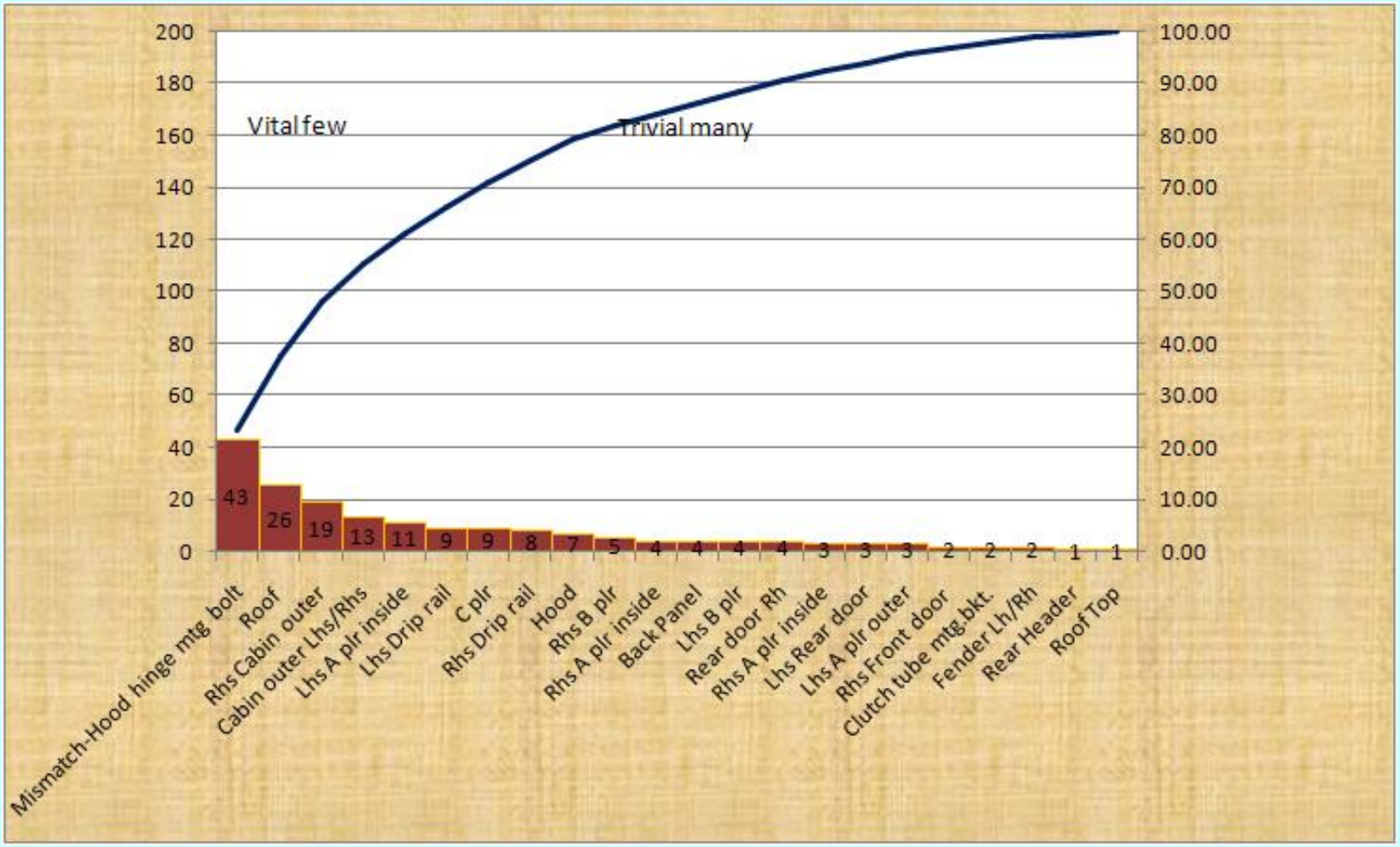
WHERE – At sill side bottom area

WHEN – During buy-off

WHO – Buy-off operator & check-man

HOW – Total 38 incidences found out of 498 in F12 FTBO

# 2. Justify the choice (reasons)




Rework Cost- 10000/annum
Severity- P1 & P2
Rework man-hrs- 8 Hrs

# 2.a Immediate Containment Action

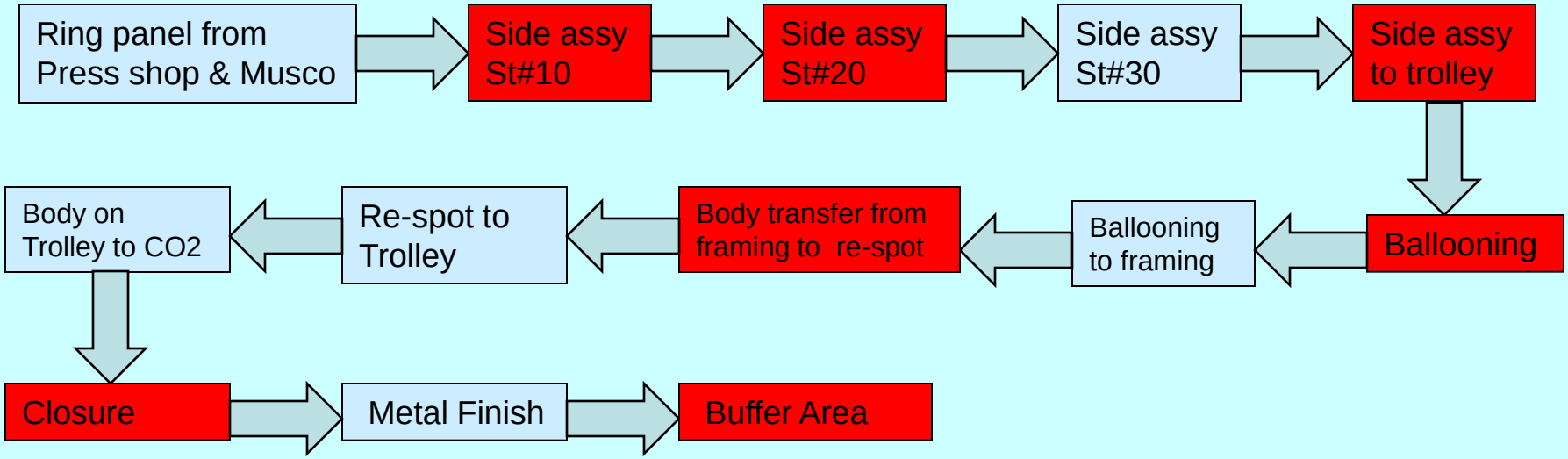
1) 100% check at buy-off

2) 100% check in check-man list

3) Reworking of damage bodies.

# 3. Understand the current situation

## Observation of process




Probable area

# 3. Understand the current situation

## Observation of process

**Rocker Panel damage has been observed all over the place of sill side**

# 3. Understand the current situation

## Data monitoring for FRC & FTBO level at SCDC

| Shift | Months | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Oct-11 | Nov-11 | Dec-11 | Jan-12 | Feb-12 | Mar-13 | Apr-12 | May-12 | Jun-12 |
| FTBO | 2 | 2 | 4 | 6 | 5 | 6 | 5 | 6 | 2 |
| FRC | 25 | 30 | 36 | 24 | 32 | 30 | 22 | 20 | 10 |

- **After scrutinizing the data we observed @ 8 % rework in FTBO & 3 to 5% rejection in FRC is due to issue of rocker panel damage.**

- **Irrespective of shift and operator skill level the concern persists in all the shifts erratically.**

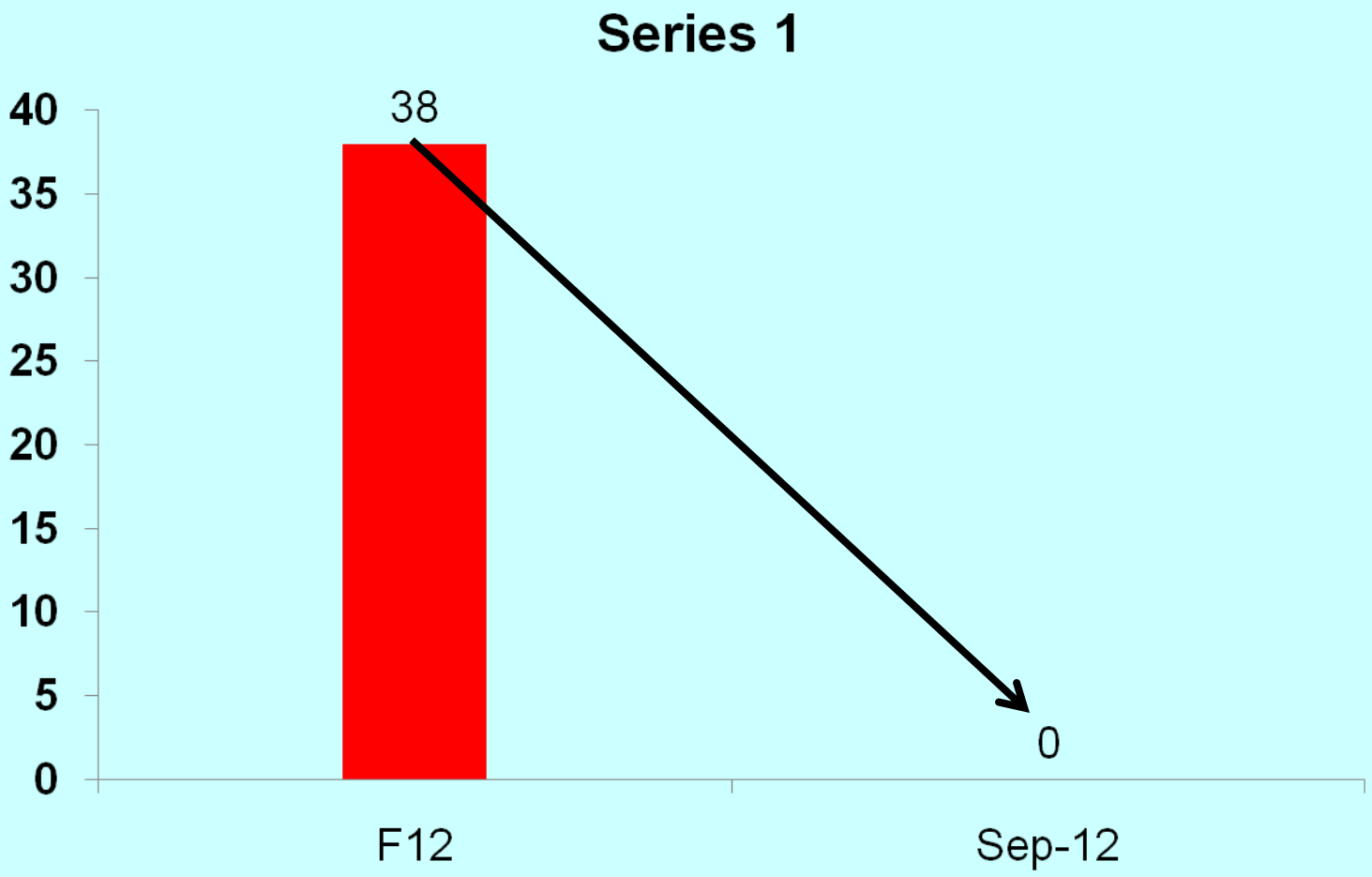
# 3. Understand the current situation

## Observation of process

- Rocker panel damage observed right from side assy.  to re-spot stages.
- Rocker panel damage also observed at metal finishing area.
- Rocker panel damage also observed during loading of bodies after process.
- Total 38 incidences found out of 498 in F12 FTBO
- In F-13 also there is a monthly avg. of 4-incidences.
- In FRC also rocker panel damage has a contribution of 3 to 5 % daily.

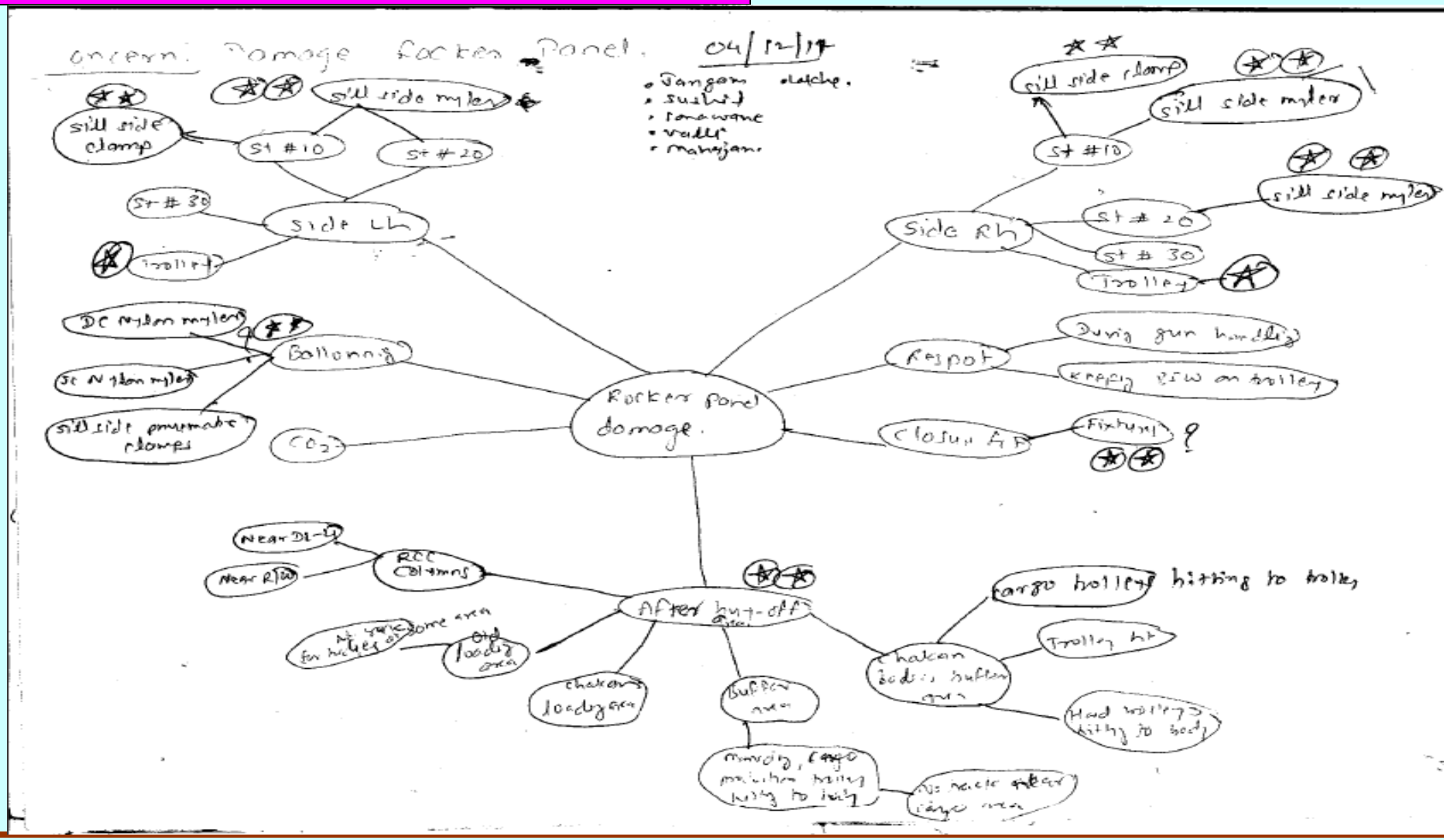
# 4. Select Target

Identify the target. A target is a number indicating the level of improvement to be achieved.



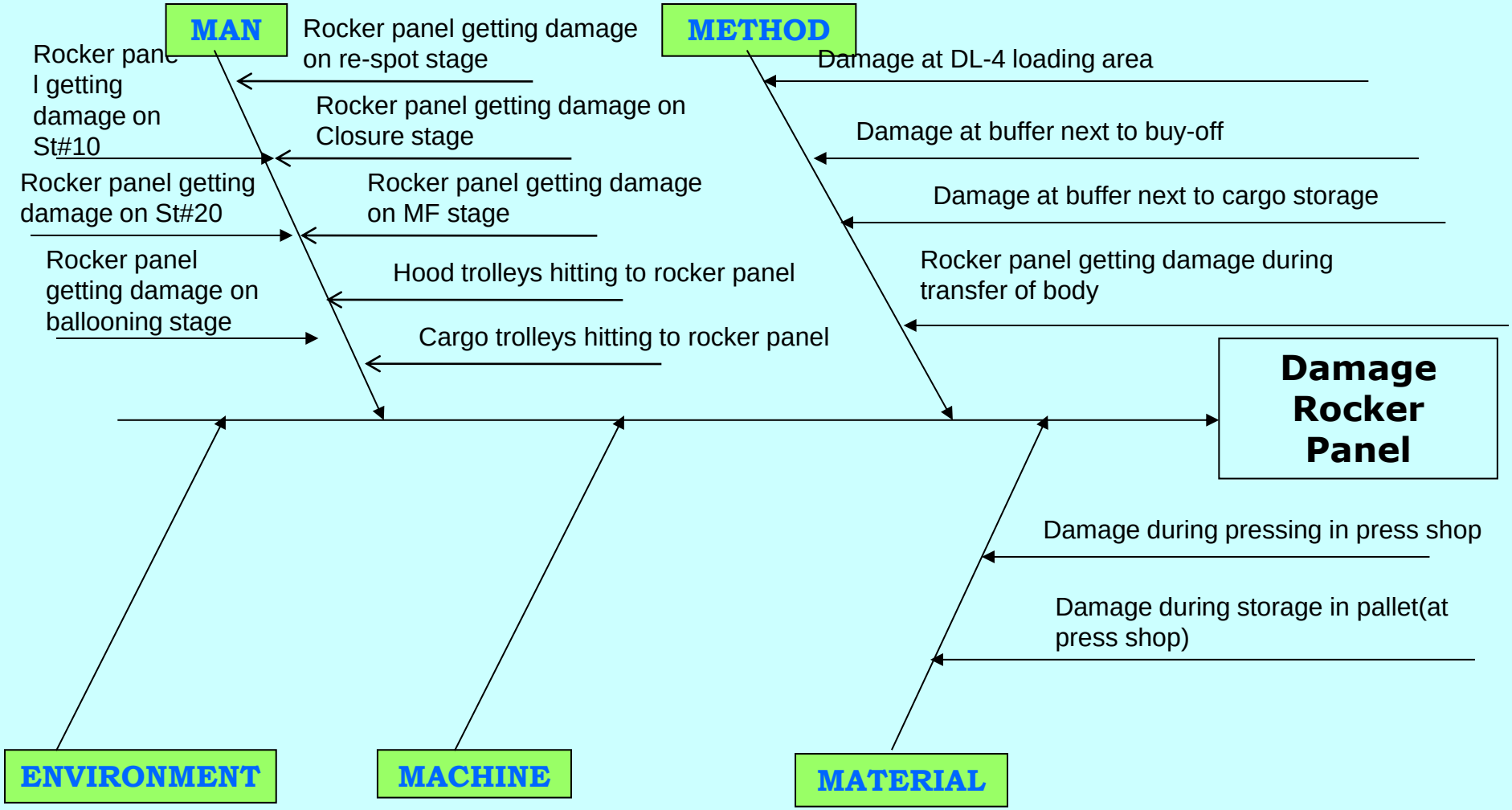
# 5. Analysis

## 5.1 Identify possible causes:



# 5. Analysis

## 5.1 Identify possible causes:



# List of possible causes

| Sr. no. | Possible Causes |
|---|---|
| 1 | Damage during storage in pallet(at press shop) |
| 2 | Rocker panel getting damage on St#10 |
| 3 | Rocker panel getting damage on St#20 |
| 4 | Rocker panel getting damage on ballooning stage |
| 5 | Rocker panel getting damage during transfer of body |
| 6 | Rocker panel getting damage on re-spot stage |
| 7 | Rocker panel getting damage on Closure stage |
| 8 | Damage at buffer next to buy-off |
| 9 | Damage at DL-4 loading area |
| 10 | Damage at buffer next to cargo storage |
| 11 | Hood trolleys hitting to rocker panel |
| 12 | Cargo trolleys hitting to rocker panel |

# TEST OF HYPOTHESIS

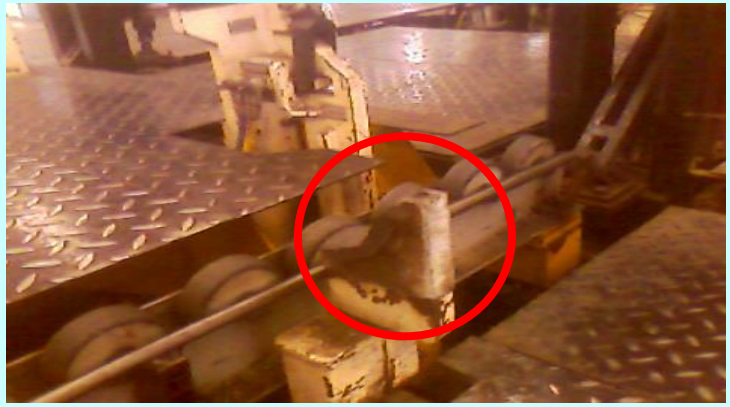
## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing and Observations | Conclusion |
|---|---|---|---|
| 1 | Rocker panel getting damage on St#10 | Operators asked to load the ring panel on st#10 ,by deliberately hitting to sill side mylers. Same damages observed as we are receiving in FRC & FTBO | **Valid** |
| 2 | Rocker panel getting damage on St#20 | Operators asked to load the ring panel on st#10 ,by deliberately hitting to sill side mylers. Same damages observed as we are receiving in FRC & FTBO | **Valid** |

# TEST OF HYPOTHESIS

## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing and Observations | Conclusion |
|---|---|---|---|
| 3 | Rocker panel getting damage on ballooning stage | Side sub-assy marked for OK before ballooning stage & immediately observed after ballooning stage. Rocker panel damage observed | Valid |



# TEST OF HYPOTHESIS

## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing and Observations | Conclusion |
|---|---|---|---|
| 4 | Rocker panel getting damage during transfer of body | Side sub-assy marked for OK at ballooning stage & immediately observed after re-spot stage. Rocker panel damage observed | Valid |

# TEST OF HYPOTHESIS

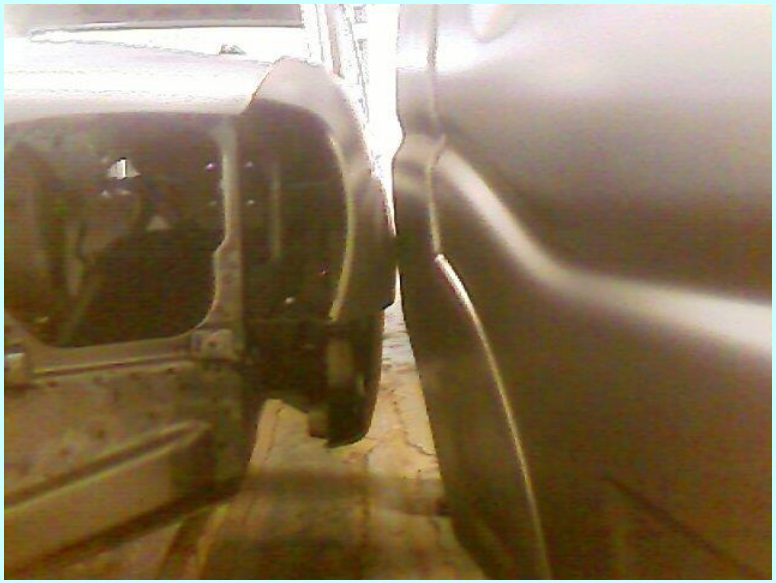
## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing and Observations | Conclusion |
|---|---|---|---|
| 5 | Rocker panel getting damage on Closure stage | 1) Rocker panel area observed during setting of fixture. Sill side lower clamps hitting to rocker panel<br>2) Side sub-assy marked for OK at CO2 stage & immediately observed after Closure stage. Rocker panel damage observed | Valid |

# TEST OF HYPOTHESIS

## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing and Observations | Conclusion |
|---|---|---|---|
| 6 | Damage at buffer next to buy-off | Bodies kept very close to each other . Rocker panel hitting from one body to another | Valid |
| 7 | Damage at DL-4 loading area | Bodies kept very close to each other . Rocker panel hitting from one body to another | Valid |
| 8 | Damage at buffer next to cargo storage | Bodies kept very close to each other . Rocker panel hitting from one body to another | Valid |



# TEST OF HYPOTHESIS

## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing and Observations | Conclusion |
|---|---|---|---|
| 9 | Hood trolleys hitting to rocker panel | Hood trolley brought near to bodies kept on trolley. It is observed that the hood trolley is inserting inside the body trolley & damaging the rocker panel | Valid |
| 10 | Cargo trolleys hitting to rocker panel | Cargo trolley brought near to bodies kept on trolley. It is observed that the cargo trolley is inserting inside the body trolley & damaging the rocker panel | Valid |

# WHY - WHY Analysis

## 5.3 Identify the root causes: (Use Format MQS/MAPS/F05)

| Valid Probable Cause | WHY | WHY | WHY | WHY | WHY |
|---|---|---|---|---|---|
| Rocker panel getting damage on St#10 | Getting damage during loading | While loading rocker panel hitting to fixture myler | Fixture mylers are of M S & having sharp edges | No insulation for mylers | |
| Rocker panel getting damage on St#20 | Getting damage during loading | While loading rocker panel hitting to fixture myler | Fixture mylers are of M S & having sharp edges | No insulation for mylers | |
| Rocker panel getting damage on ballooning stage | Getting damage during loading | While loading rocker panel hitting to body support blocks | Body support blocks are of M S & having sharp edges | Hard material for body support blocks | |
| Rocker panel getting damage during transfer of body | Rocker panel hitting to vertical rollers | Vertical rollers are of Teflon | Hard material for vertical rollers | | |
| Rocker panel getting damage on Closure stage | Closure fixture lower clamps hitting to rocker panel | Closure fixture lower clamps are of M S & having sharp edges | Closure fixture lower clamps are of M S & having sharp edges | No insulation for Closure clamps | |

# WHY - WHY Analysis

## 5.3 Identify the root causes: (Use Format MQS/MAPS/F05)

| Valid Probable Cause | WHY | WHY | WHY | WHY | WHY |
|---|---|---|---|---|---|
| Damage at buffer next to buy-off | **Bodies are kept very close to each other** | **No separation facility between two bodies** | | | |
| Damage at DL-4 loading area | **Bodies are kept very close to each other** | **No separation facility between two bodies** | | | |
| Damage at buffer next to cargo storage | **Bodies are kept very close to each other** | **No separation facility between two bodies** | | | |
| Hood trolleys hitting to rocker panel | **Hood trolley hitting rocker panel during transportation** | **Hood trolley entering inside the body on trolley** | **No restriction for hood trolley** | | |
| Cargo trolleys hitting to rocker panel | **Cargo trolley hitting rocker panel during transportation** | **Cargo trolley entering inside the body on trolley** | **No restriction for cargo trolley** | | |

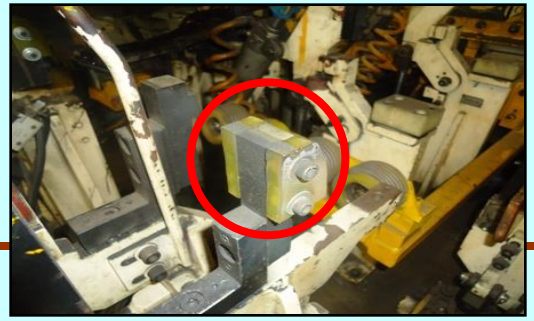
# 6. Implement the corrective measures

## 6.1 Actions Decided

| Root Cause | Action Plan | Target Dt | Resp. | Status |
|---|---|---|---|---|
| **No insulation for mylers on St#10** | 1) Milers (10nos Lh / Rh) to be insulated by PU material as per verito body-shop<br>2) Upper clamps(8nos Lh / Rh) to be protected by rubber rollers | 30.05.12 | Jangam | Close |
| **No insulation for mylers on St#20** | 1) Milers to be insulated by PU material as per verito body-shop | 30.05.12 | Jangam | Close |

**BEFORE**

**AFTER**



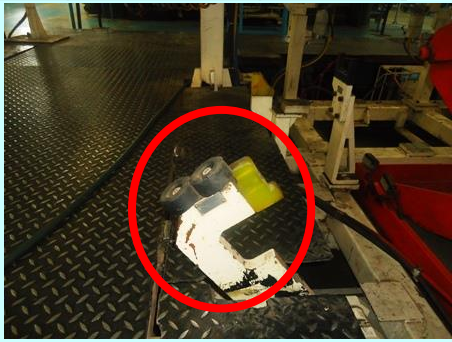
# 6. Implement the corrective measures

## 6.1 Actions Decided

| Root Cause | Action Plan | Target Dt | Resp. | Status |
|---|---|---|---|---|
| **Hard material for body support blocks at ballooning** | Hard material to be replace with PU material blocks(6nos Lh/Rh) | 25.05.12 | Jangam | Close |

**BEFORE**

**AFTER**



# 6. Implement the corrective measures

## 6.1 Actions Decided

| Root Cause | Action Plan | Target Dt | Resp. | Status |
|---|---|---|---|---|
| **Hard material Hard material for vertical rollers** | Taflon rollers to be replace with soft material rollers(12nos Lh/Rh) | 28.05.12 | Jangam | Close |

**BEFORE**

**AFTER**

# 6. Implement the corrective measures

## 6.1 Actions Decided

| Root Cause | Action Plan | Target Dt | Resp. | Status |
|---|---|---|---|---|
| **No insulation for Closure clamps** | Closure lower clamps to be insulated by rubber hose pipe | 20.04.12 | Sushil | Close |

**BEFORE**

No insulation for Lower clamps

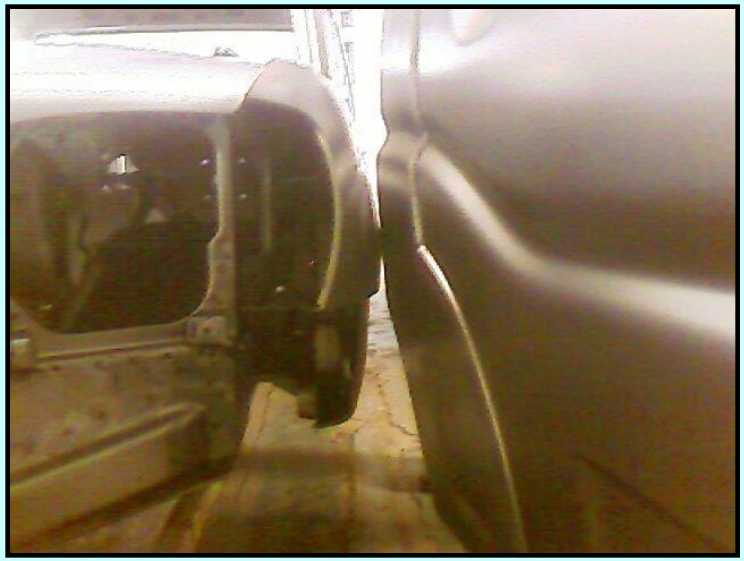
**AFTER**

# 6. Implement the corrective measures

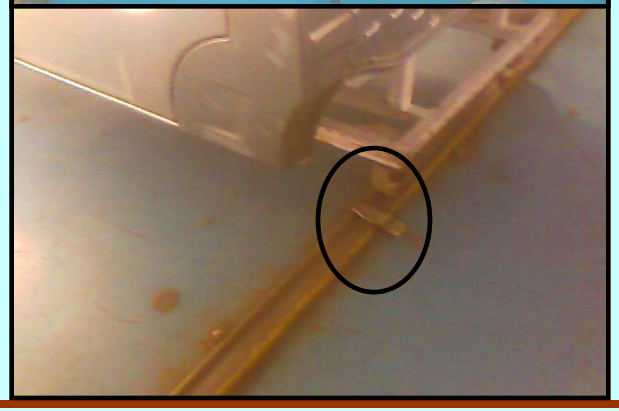
## 6.1 Actions Decided

| Root Cause | Action Plan | Target Dt | Resp. | Status |
|---|---|---|---|---|
| **No separation facility between two bodies at buffer next to buy-off** | 1) Body side guards provided (41 pairs)<br>2) Stopper provided in between 2-bodies<br>3) Body buffer reduced by 1no.(4 bodies to be kept as against 5) | 30.07.12 | Jangam /Rajesh | Close |
| **No separation facility between two bodies at DL-4 loading area** | 1) Body side guards provided (41 pairs)<br>2) Body track provided(2-nos)<br>3) Body buffer reduced by 1no.(4 bodies to be kept as against 5) | 30.07.12 | Jangam | Close |
| **No separation facility between two bodies at buffer next to cargo storage** | 1) Body side guards provided (41 pairs)<br>2) Body track provided(1-nos)<br>3) Body buffer reduced by 3no.(6 bodies to be kept as against 9) | 30.07.12 | Jangam | Close |

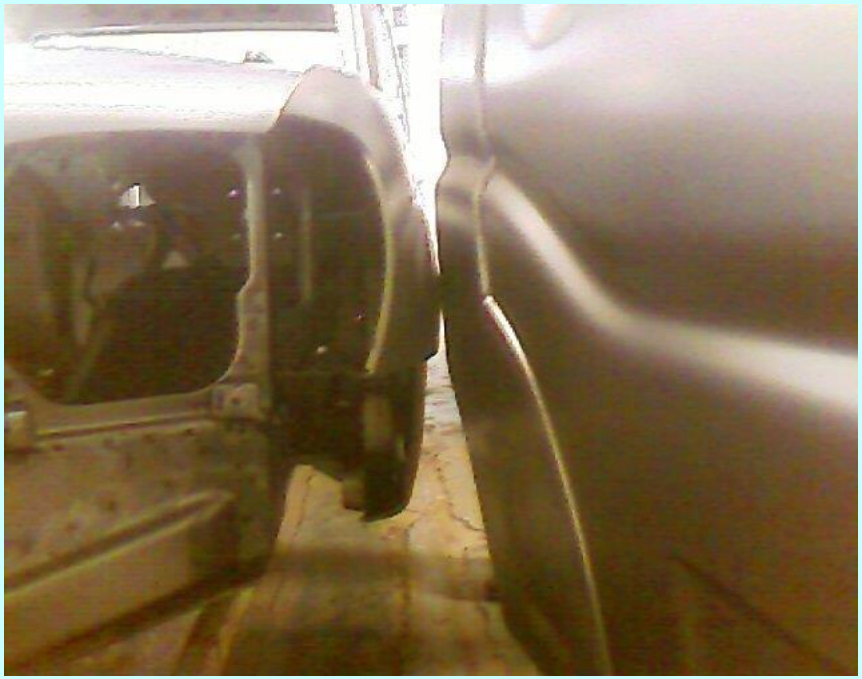
# BEFORE



# AFTER



# BEFORE



# AFTER

BACK BACK-1

# BEFORE

# AFTER

BACK BACK-1

# 6. Implement the corrective measures

## 6.1 Actions Decided

| Root Cause | Action Plan | Target Dt | Resp. | Status |
|---|---|---|---|---|
| **No restriction for hood trolley** | Vertical stoppers to be provided to hood trolley on lower side so that trolleys will not enter inside the body on trolley | 22.06.12 | Jangam /Rajesh | Close |

**BEFORE**

**AFTER**

# 6. Implement the corrective measures

## 6.1 Actions Decided

| Root Cause | Action Plan | Target Dt | Resp. | Status |
|---|---|---|---|---|
| **No restriction for cargo trolley** | Vertical stoppers to be provided to cargo trolley(21nos) on lower side so that trolleys will not enter inside the body on trolley | 22.06.12 | Jangam /Rajesh | Close |

**BEFORE**

**AFTER**

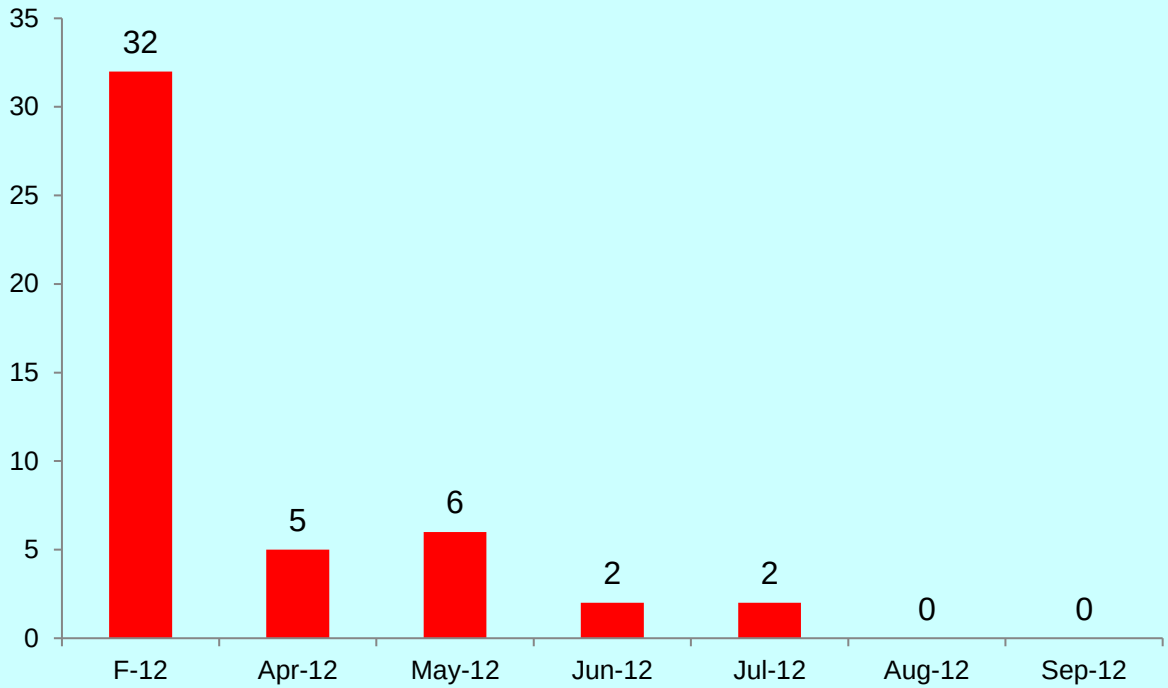
# BEFORE

# AFTER

BACK BACK-1

# 7. Check & confirm the results & effects

## 7.1 Verify Results:




## 7.2 Full Scale Implementation plan:

All actions are closed

# Check & confirm the results & effects (Cont……)

## 7.3 Tangible & Intangible benefits:

Rework man-hrs saved- 18hrs

Rework cost saved – 12000/annum

Operator irritant removed during extreme lower side rework

# 8. Standardisation

## 8.1 Updation of documents:

1) Fixture History card
2) Control Plan
3) PFMEA

## 8.2 Training:

1) Trg given to casual labours for fitting & removing of side body guards.

## 8.3 Additional Audit / Check

1)Fixture PM sheet updated for PU insulation of mylers
2) Fixture PM sheet updated for rubber roller insulation of clamps

# 9. Summarise & Plan future actions

## 9.1 Conclusion:

For any concern resolution if we go through process flow & explore each & every possible area, then there is a 100% possibility of eliminating the concern.

## 9.2 Horizontal Deployment:

1) RH side actions horizontally deployed to LH side

## 9.3 Lessons Learnt:

Ding/dent, damages are to be taken in consideration at least at the stage of PTR with the experience of shop floor people

**Thank You……MQS TEAM**